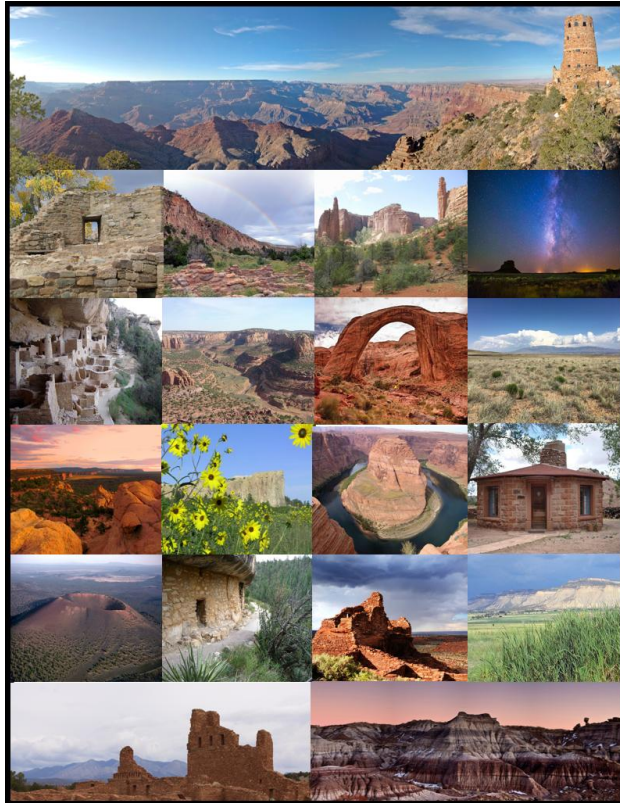




# Data Life Cycles and Workflows for the Southern Colorado Plateau Network Vital Signs

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### Data Life Cycles and Workflows for the Southern Colorado Plateau Network (SCPN) Vital Signs

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## 1 Introduction

This document outlines the data life cycles for the eight Southern Colorado Plateau Network (SCPN) vital signs as implemented through six monitoring protocols within the National Park Service's Inventory and Monitoring Division.

Because data collection, management, quality assurance, and reporting are structured operationally by monitoring protocol, this document is organized by protocol, with associated vital signs identified for each.

The monitoring protocols and associated vital signs monitored by SCPN include [Monitoring Protocol (Vital Sign(s))]:

1. Riparian (Alluvial Groundwater and Water Surface Flow)
2. Aquatic Macroinvertebrate (Aquatic Macroinvertebrates)
3. Water Quality (Stream Water Quality).
4. Integrated Upland (Upland Vegetation and Soil Aggregate Stability)
5. Landbird Monitoring Protocol Implementation Plan for Park Units in the Southern Colorado Plateau Network (Landbirds).
6. Satellite-based Vegetation Phenology and Condition and Snow Cover Extent Monitoring Protocol for the Northern and Southern Colorado Plateau Networks (Land Surface Phenology).

The SCPN also supports three automated climate monitoring stations that were inherited by parks (NAVA, PEFO, WUPA) after stations were abandoned when NOAA discontinued the Regional Climate Reference Network program.

The purpose of this guide is to support consistent and transparent data management practices across the network. It is intended as a working document for field staff, data managers, and program leads, serving both as a reference for current workflows and a tool to identify future needs and areas for improvement. Each section of this document describes the ideal data life cycle for a given monitoring protocol, identifies roles and responsibilities for each task, and summarizes current operation status of those tasks as of December 2025.

Tasks are grouped by type:

- Data collection and preparation
- Data processing
- Data analysis, reporting, and publication

Each task is also assigned a status:

## Introduction

- **Operational:** Task is currently functioning but may require ongoing maintenance or minor updates.
- **In progress:** Task is actively underway with resources assigned; completion planned in current cycle
- **Needs to be updated:** Task requires a major update or rebuild for lifecycle to continue; update is not yet possible until one or more 'In progress' tasks are complete.
- **Outstanding:** Task is not yet implemented and requires the most development.
- **Highly variable (Lab):** Task is subject to external variability or delays, often associated with laboratory timelines.
- **Future development:** Task is intended to improve efficiency and cannot be completed until 'In progress' tasks are complete.

Finally, each task is labeled with the **role(s)** responsible for its completion:

- **PL** = Project Lead
- **DM** = Data Manager
- **FCL** = Field Crew Lead
- **DMA** = Data Manager Assistant
- **GIS** = GIS Specialist

The table below summarizes roles and responsibilities across core monitoring tasks using the RACI framework:

1. **Responsible** (does the work)
2. **Accountable** (owns the outcome)
3. **Consulted** (provides input)
4. **Informed** (kept updated).

| Task / Activity         | Responsible | Accountable | Consulted | Informed |
|-------------------------|-------------|-------------|-----------|----------|
| Field Data Collection   | FCL         | PL          | DMA, DM   | PM       |
| Data Entry / Processing | FCL         | PL          | DMA, DM   | PM       |
| QA/QC                   | PL          | PL          | DM, FCL   | PM       |
| Data Storage & Security | DM          | PL          | DMA, GIS  | PM       |
| Analysis & Reporting    | PL          | PL          | DM, GIS   | PM       |
| Data Publication        | DM          | PL          | -         | PM       |

## 2 Water Resources

SCPn currently has four vital signs that fall under the water resources category: Alluvial Groundwater, Water Surface Flow, Aquatic Macroinvertebrates, and Water Quality. Data are gathered and managed for these vital signs under three monitoring protocols (Riparian, Aquatic Macroinvertebrates, and Water Quality) with data management at various stages of completion across protocols.

### 2.1 Riparian (Alluvial Groundwater and Water Surface Flow)

#### 2.1.1 Ideal Data Life Cycle for the Riparian protocol (Alluvial Groundwater and Water Surface Flow)

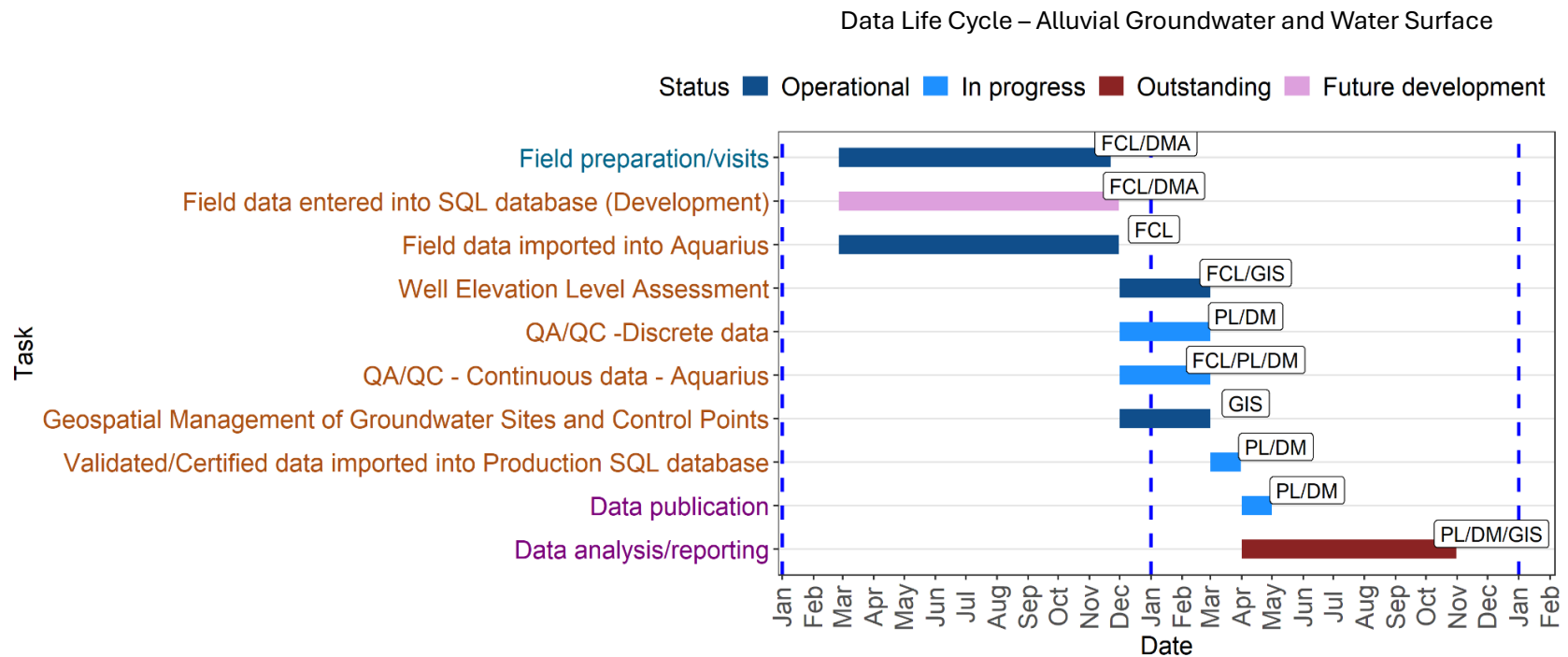


Figure 1.1 This chart is a representation of the workflow for the Alluvial Groundwater and Water Surface Flow vital signs. Tasks are color coded according to the type of task, where data collection tasks (blue text), data entry and QA/Qc tasks (orange text), and data analysis, reporting, and publication tasks (purple text). Bars for each task are color coded according to status of the task. 'Operational' (dark blue): task

is currently functioning but may require ongoing maintenance or minor updates. *'In progress'* (light blue): work is actively underway with resources assigned; completion planned in current cycle. *'Outstanding'* (red): Task is not yet implemented and requires the most development. *'Future development'* (light purple): task is intended to improve efficiency and cannot be completed until 'In progress' tasks are complete. Each task bar is labeled with the person(s) that play a role in completing the task, where FCL = Field Crew Lead, DMA = Data Manager Assistant, PL = Project Lead, and DM = Data Manager.

## 2.1.2 Task descriptions and Roles and Responsibilities

### 2.1.2.1 *Data collection and preparation*

#### 2.1.2.1.1 Field preparation/visits – Field crew lead and assistant data manager

Field preparation/visits take place between late-February through the end of November. Data Management field preparation consists of incorporating any changes, fixes, or updates to the Access field data entry application. The hydrologic technician packs all sampling gear, ensures there are adequate spare Levelloggers/Barologgers, and reviews the prior trip report for each park to ensure no gear is forgotten.

During field visits, the hydrologic technician retrieves logger data from Solinst Levelloggers and Barologgers, manually measures the distance from the top of the well (Alluvial Groundwater) or gauge (Water Surface Flow) to the water, and performs a rapid well elevation level assessment using a SOKKIA Auto Level to ensure well elevations have not changed since the last high-accuracy level loop survey. Discrete field data are recorded on water-resistant paper datasheets in the field. Although water level loggers collect data continuously, periodic site visits are necessary to retrieve data and inspect equipment. The operational goal is four visits per year when feasible, with scheduling prioritized around Aquatic Macroinvertebrate and Water Quality monitoring windows. Attempts to reduce visits to twice annually were not sufficient to address equipment maintenance.

### 2.1.2.2 *Data processing*

#### 2.1.2.2.1 Discrete data entered into SQL database – Field crew lead and assistant data manager

Discrete data are entered into the SCPN\_Groundwater\_DataEntry database on the Development instance of SQL server after each field visit. This is performed using the Access Groundwater Application, which allows the user to enter data from paper datasheets into SQL. This is primarily done by the field crew lead with help from the assistant data manager. Future development will allow for this application to be loaded onto a field tablet, so discrete data can be entered while in the field. Alluvial groundwater and water surface flow discrete data are stored in the same database on SQL, so most of the discrete data is shared between these two vital signs.

#### 2.1.2.2.2 Continuous data imported into Aquarius – Field crew lead

Continuous data collected by the field loggers are uploaded into Aquarius (<https://irma.nps.gov/aqwebportal/>) after each field visit by the field crew lead. Discrete data (manual depth to water) will be entered into Aquarius at this time as well.

#### 2.1.2.2.3 Well Elevation Level Assessment – Field crew lead and GIS Specialist

Each site undergoes an annual rapid assessment of well elevation levels to monitor for any vertical shifts over time. After the field visit, the hydrologic technician scans the Rapid Well Elevation Level datasheets and saves the digital copy into the designated field datasheet file structure. The GIS specialist enters these data into a Rapid Well Elevation Level database in Access and then confirms that the elevation assessment data falls within the acceptable error margin of +/- 1 centimeter. If any well exceeds this threshold, the GIS specialist conducts a resurvey using a high-accuracy level loops survey technique to ensure precision.

#### 2.1.2.2.4 QA/QC and validation of discrete data – Project Lead and data manager

QA/QC of the discrete data should be conducted once a year. In 2025, the QA/QC script was updated by the data strike team and is currently under review the project lead and the data manager. Moving forward, the project lead will run the script and confer with data manager for any corrections that are required. Once the project lead and data manager have confirmed all corrections, the data manager will validate the dataset by ensuring all errors have been resolved and then upload the validated data to the *SCPN\_Groundwater* database in the Production instance of the SQL Server. If QA/QC scripts indicate changes need to be made to discrete data in Aquarius, the field crew lead will make these corrections.

#### 2.1.2.2.5 QA/QC continuous data – Project Lead, field crew lead, and data manager

QA/QC of the continuous data should be conducted throughout the year as data comes in. The crew lead provides preliminary grades for the data in Aquarius, which are later approved by the project lead and data manager.

#### 2.1.2.2.6 Geospatial Management of Sites and Control Points – GIS Specialist

The GIS specialist is responsible for managing the geospatial site locations for groundwater data and control points used in a survey database. This includes maintaining an updated GIS database that contains all relevant spatial data, including well locations and elevations, control points, and metadata. The GIS specialist ensures that these geospatial datasets are accurate, consistently formatted, and integrated with other data elements in the groundwater management workflows. Any changes to site locations or survey control points are documented and updated in the GIS system to ensure proper data interpretation and analysis.



### 2.1.2.3 *Data analysis, reporting, and publication*

#### 2.1.2.3.1 *Data publication – Project Lead/data manager*

Data publication of discrete data will occur after data has undergone QA/QC and is validated by the data manager. Metadata for this protocol will need to be updated and formatted into EML. The data manager will collaborate with the Project Lead to ensure the final data package is properly formatted using EML and that all data files are in a machine-readable format (CSV), as provided by the project lead.

The data manager will certify the data by ensuring that the data package has been verified using NPS provided tools and that data files are the same as data in the Production instance of SQL server, excluding any formatting or protected data. The data manager will publish the data package on NPS Data Store after review from the Project Lead and participating parks.

Continuous and discrete data in Aquarius will be released for public view after the project lead and data manager have approved the grades.

#### 2.1.2.3.2 *Data analysis/reporting – Project Lead/data manager, and GIS Specialist*

We should discuss what types of reports are expected moving forward, but at the minimum, data summaries should be completed. Further discussions should be had regarding trend reports. The GIS Specialist performs spatial analyses as needed to support reporting efforts and develops maps for annual reporting. Additionally, they review annual summaries and reports to ensure the accuracy of spatial data and information.

### 2.1.3 *Brief status update*

Alluvial Groundwater has been collected since 2007 and is still currently being collected in 2024.

We are less confident in the initial survey measurements for the well elevations (m), an essential value needed to properly calculate groundwater levels. As of November 2019, a new survey was completed to properly obtain these well elevations. Further, other discrete measurements made during this time (such as measure point to ground, measure point to water level) were taken with various measurement tools, which resulted in data being collected in feet and in meters. Views that normalize this variation in field measurement have already been created for the SQL Server database.

Validated discrete data are stored in the Production instance of SQL server. While improvements could be made to the database, it is adequate for housing these data. As a work around to the current database structure, we will create views or R scripts to wrangle these data into tables that can be used for data publication. There are currently some metadata in the database that needs to be

updated. A table/view should be created that compiles this information. Metadata needs to be updated to EML metadata standards. QA/QC has been performed on the data up to 2018. A new script will need to be written to QA/QC more recently collected data.

## 2.2 Water Quality

### 2.2.1 Ideal Data Life Cycle for Water Quality

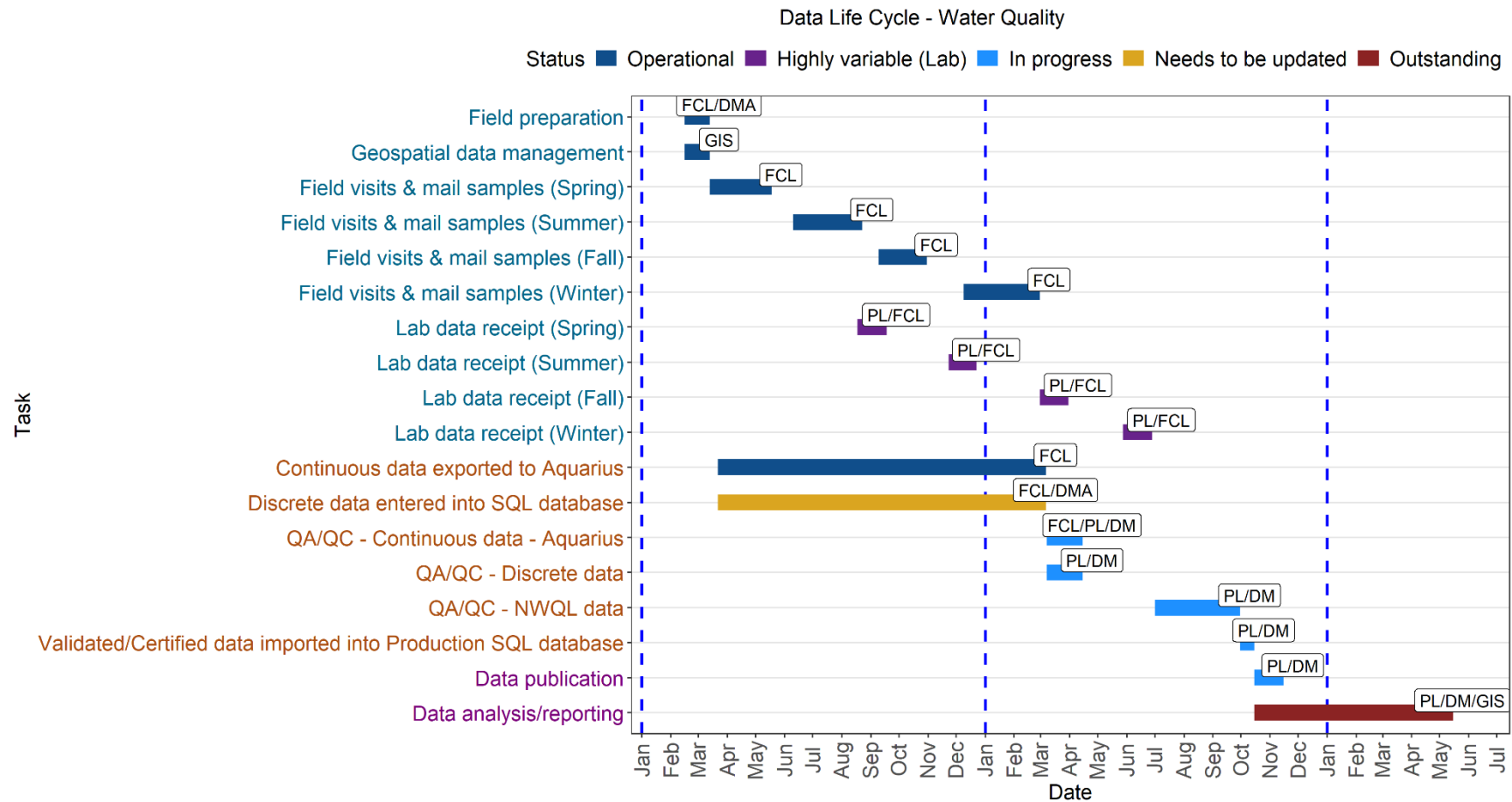


Figure 1.2. This chart is a representation of the workflow for the water quality vital sign. Tasks are colored coded according to the type of task, where data collection tasks ( blue text), data entry and QA/Qc tasks (orange text), and data analysis, reporting, and publication tasks

(purple text). Bars for each task are color coded according to status of the task. *'Operational'* (dark blue): task is currently functioning but may require ongoing maintenance or minor updates. *Highly variable (Lab)*: Task is subject to external variability or delays, often associated with laboratory timelines. *'In progress'* (light blue): work is actively underway with resources assigned; completion planned in current cycle. *'Outstanding'* (red): Task is not yet implemented and requires the most development. *'Future development'* (light purple): task is intended to improve efficiency and cannot be completed until 'In progress' tasks are complete. Each task bar is labeled with the person(s) that play a role in completing the task, where FCL = Field Crew Lead, DMA = Data Manager Assistant, PL = Project Lead, and DM = Data Manager.

## 2.2.2 Task descriptions and Roles and Responsibilities

### 2.2.2.1 Data collection and preparation

#### 2.2.2.1.1 Field preparation/visits – Field crew lead and assistant data manager

Field preparation/visits take place four times throughout the year, where samples are taken in the spring, summer, fall, and winter. Data Management field preparation consists of incorporating any changes, fixes, or updates to the Access field data entry application. The hydrologic technician packs all water quality sampling gear, ensures there are adequate spare loggers (water/air temperature and Sonde loggers), and reviews the prior trip report for each park to ensure no gear is forgotten.

During field visits, the hydrologic technician retrieves logger data from Multi-parameter Sonde and temperature loggers (water and air), records discrete data on paper datasheets (stream channel and flow conditions, stream discharge measured with a flow meter), and collects water samples. Separate water samples are collected for water chemistry, bacteria, and turbidity. Water chemistry samples are shipped to the laboratory for processing. Results from the lab are emailed to the project lead, data manager, and field crew lead and are saved in their original format to the common drive server by the project lead. Water samples for bacteria and turbidity are processed in the field by the hydrologic technician, and the results (discrete data) are recorded on paper datasheets. Sonde logger files are converted to excel files and placed in the common drive by the hydrologic technician.

#### 2.2.2.1.2 Geospatial data management – GIS Specialist

Water quality site locations are managed within an ESRI file geodatabase environment by the GIS Specialist. The GIS Specialist updates site location information as needed and provides field maps and GPS units containing site coordinates to ensure efficient navigation during fieldwork. They maintain location attribute information for each site, along with metadata documenting spatial data development processes to uphold data integrity and accessibility. QA/QC checks and visual inspections are conducted to validate the accuracy of the spatial data.

#### 2.2.2.2 *Data processing*

##### 2.2.2.2.1 Field data entered into SQL database – Field crew lead and assistant data manager

After fourth quarter sampling, the discrete field data are entered into the Water Quality database using the Access Water Quality Application, which allows the user to enter data from paper data sheets into an Access database. This is done by the field crew lead with help from the assistant data manager. After entering data, data are verified by an additional field crew member. Future development of a field tablet-compatible Access data entry application will allow the field technician to enter all discrete data in the field. Data will then be uploaded into the Development instance of SQL server for eventual QA/QC.

##### 2.2.2.2.2 Field data imported into Aquarius – Field crew lead

Continuous and discrete data collected for water and air temperature are uploaded onto Aquarius after each field visit by the hydrologic technician.

##### 2.2.2.2.3 QA/QC and validation of field (discrete) data – Project Lead and data manager

QA/QC of the discrete data should be conducted once a year. Before QA/QC, excel files for sonde logger and water chemistry files are formatted to be incorporated into the SQL database. This is the primary function of the project lead, but the data manager will help with troubleshooting and creating the R scripts. Once all files are processed, QA/QC can begin.

We are currently establishing a workflow for completing QA/QC. A formal R script will need to be written for this protocol for future QA/QC. The project lead will work with the data manager to determine what QA/QC checks need to be done on the data and for assistance with developing the QA/QC R script. Once the script is created, this will become the primary responsibility for the project lead, although the data manager will assist with any troubleshooting. After the project lead has completed QA/QC of the data, the data manager will validate the data by ensuring all errors have been accounted for and will upload the validated data into the SCPN\_WaterQuality database on the Production instance of SQL server.

##### 2.2.2.2.4 QA/QC and validation of continuous data – Project Lead, field crew lead, and data manager

QA/QC of the continuous data is completed after each field visit. The hydrologic technician uploads continuous data and discrete water temperature data into Aquarius and grades that period of the record. The hydrologic technician completes this step with assistance from the project lead and data manager as needed.

### *2.2.2.3 Data analysis, reporting, and publication*

#### *2.2.2.3.1 Data publication – Project Lead and data manager*

Data publication of discrete data will occur after data has undergone QA/QC and is validated by the data manager. Metadata for this protocol will need to be updated and formatted into EML. The data manager will collaborate with the project lead to ensure the final data package is properly formatted using EML and that all data files are in a machine-readable format (CSV), as provided by the project lead.

The data manager will certify the data by ensuring that the data package has been verified using NPS provided tools and that data files are the same as data in the Production instance of SQL server, excluding any formatting or protected data. The data manager will publish the data package on NPS Data Store after review from the project lead and participating parks.

Continuous and discrete data in Aquarius will be released for public view after the project lead and data manager have approved the grades.

#### *2.2.2.3.2 Data analysis/reporting – Project Lead, data manager, and GIS Specialist*

Exceedance value reporting to parks will occur annually beginning FY26. Further discussions should be had regarding trend reports. The GIS Specialist performs spatial analyses as needed to support reporting efforts and develops maps for annual reporting. Additionally, they review annual summaries and reports to ensure the accuracy of spatial data and information.

## 2.3 Aquatic Macroinvertebrates

### 2.3.1 Ideal Data Life Cycle for Aquatic Macroinvertebrates

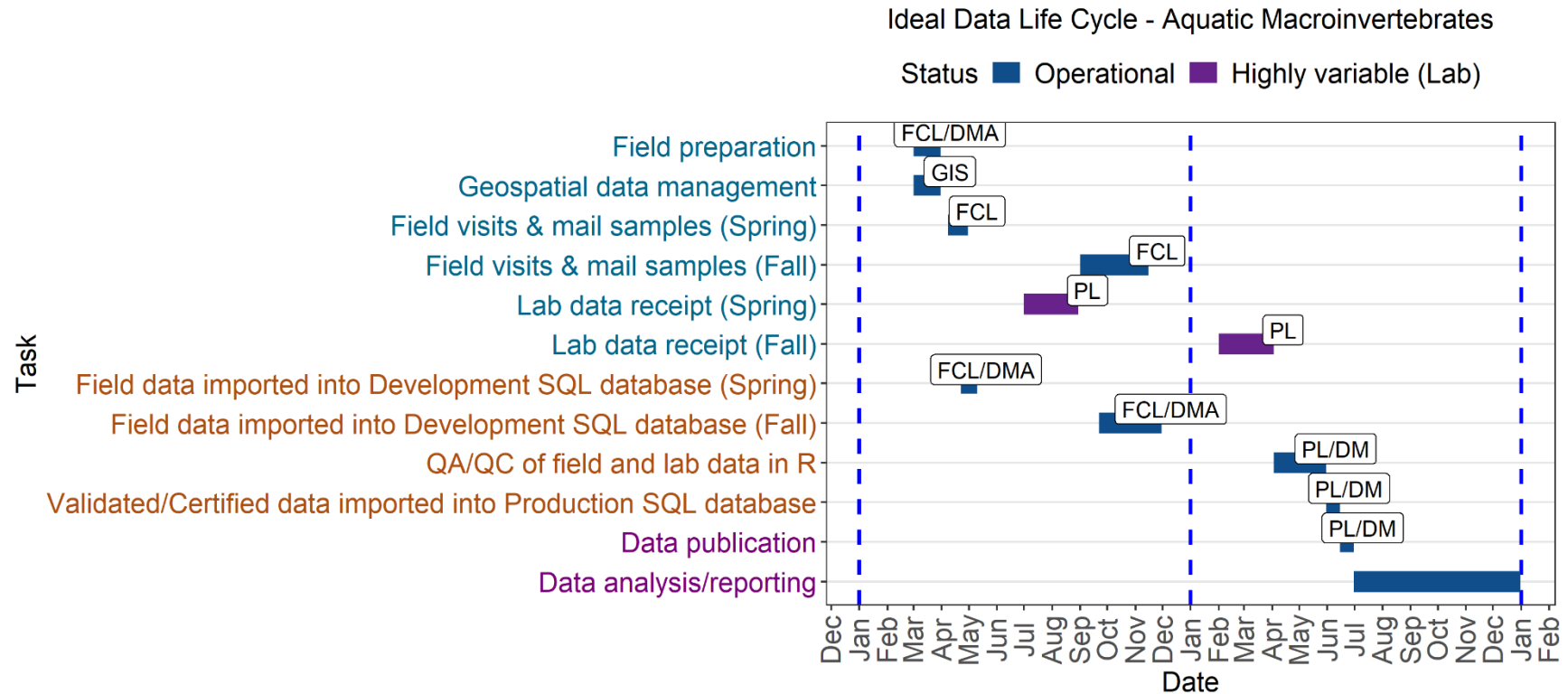


Figure 1.3. This chart is a representation of the workflow for the Aquatic Macroinvertebrates vital sign. Tasks are colored coded according to the type of task, where data collection tasks (blue text), data entry and QA/QC tasks (orange text), and data analysis, reporting, and publication tasks (purple text). Bars for each task are color coded according to status of the task. 'Operational' (dark blue): task is currently functioning but may require ongoing maintenance or minor updates. 'Highly variable (Lab)': Task is subject to external variability or delays, often associated with laboratory timelines. Each task bar is labeled with the person(s) that play a role in completing the task, where FCL = Field Crew Lead, DMA = Data Manager Assistant, PL = Project Lead, and DM = Data Manager.

## 2.3.2 Task descriptions and Roles and Responsibilities

### 2.3.2.1 *Data collection and preparation*

#### 2.3.2.1.1 Field preparation/visits – Field crew lead and assistant data manager

Field preparation/visits take place once per year at each site. Data Management field preparation consists of incorporating any changes, fixes, or updates to the Access field data entry application. The hydrologic technician packs all sampling gear required, ensures there are adequate spare loggers (temperature and Sonde loggers), and reviews the prior trip report for each park to ensure no gear is forgotten. During field visits, the hydrologic technician records various discrete data into the field data entry Access application, retrieves logger data from Multi-parameter Sonde and temperature loggers (water and air), and collects aquatic macroinvertebrate samples to ship to the lab.

#### 2.3.2.1.2 Geospatial data management – GIS Specialist

Aquatic macroinvertebrate site locations are managed by the GIS Specialist within an ESRI file geodatabase. The GIS Specialist is responsible for regularly updating site location information as needed, providing field maps, and supplying GPS units with site coordinates to ensure efficient navigation during fieldwork. They maintain location attribute information for each site and develop appropriate metadata documenting the spatial data development processes to uphold data integrity and accessibility, conducting QA/QC checks and visual inspection to validate the accuracy of the spatial data.

### 2.3.2.2 *Data processing*

#### 2.3.2.2.1 Field data entered into SQL database – Field crew lead and assistant data manager

Discrete field data are entered into the AquaticMacro Access Field Entry Application during each field visit. After each field visit, the back-end database is exported to the common drive server for safe keeping and eventually appended to the Development instance of SQL Server. Paper field datasheets are also available if the field tablets are not operable, and the data later entered in the office. Data entry is done by the hydrologic technician with help from the assistant data manager. Future development will allow for the transfer of data from field tablets to an Access AquaticMacro Application to be loaded into a Development instance of SQL server.

#### 2.3.2.2.2 QA/QC and validation of field (discrete) data – Project Lead and data manager

QA/QC of the discrete data should be conducted once a year. An R script has been written for this protocol to perform QA/QC of the data. QA/QC is the primary responsibility of the Project Lead, although the data manager will assist with any troubleshooting and importing AM files. After the Project Lead has completed QA/QC of the data, the data manager will validate the data by ensuring all

errors have been accounted for and will upload the validated data into the SCPN\_AquaticMacro database on the Production instance of SQL server.

### *2.3.2.3 Data analysis, reporting and publication*

#### *2.3.2.3.1 Data publication – Project Lead/data manager*

Data publication will occur after data has undergone QA/QC and is validated by the data manager. Metadata for this protocol has been updated and formatted into EML. The data manager will work with the Project Lead to ensure the final data package is properly formatted using EML and data files are in machine-readable format (CSV), which will be provided by the Project Lead. The data manager will certify the data by ensuring that the data package has been verified using NPS provided tools and that data files are the same as data in the Production instance on SQL server, excluding any formatting or protected data. The data manager will publish the data package on NPS Data Store after review from the Project Lead and participating parks.

#### *2.3.2.3.2 Data analysis/reporting – Project Lead/data manager, and GIS Specialist*

We should discuss what types of reports are expected moving forward, but at the minimum, data summaries should be completed. Further discussions should be had regarding trend reports. The GIS Specialist performs spatial analyses to support reporting efforts and develops maps for annual reporting. They review summaries and reports to verify the accuracy of spatial data and conduct QA/QC assessments on reported spatial information. Additionally, the GIS Specialist sources and compiles supplementary spatial data—such as burn severity data, national hydrography data, digital elevation data, and local water use data—to enhance analyses and reporting.



### 3 Birds

#### 3.1 Ideal Data Life Cycle for Birds

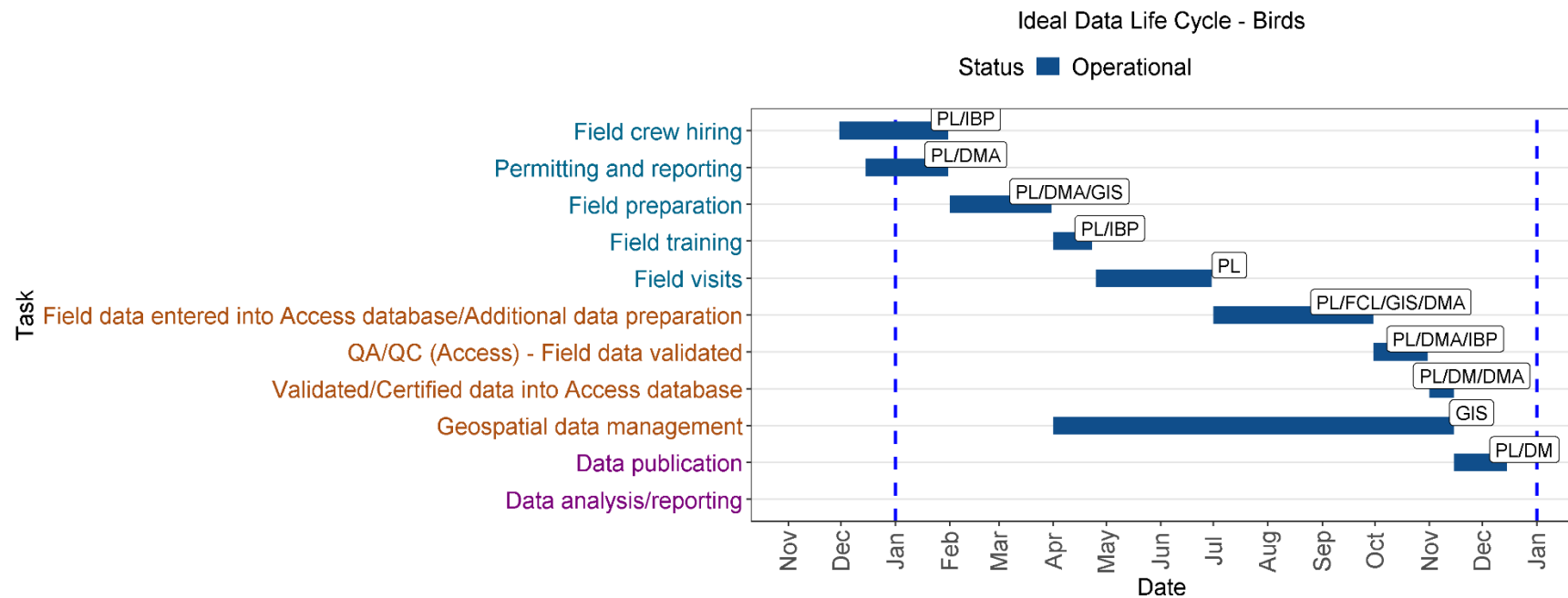


Figure 2.1 This chart is a representation of the workflow for the Birds vital sign. Tasks are colored coded according to the type of task, where data collection tasks are in blue text, data entry and QA/QC tasks are in orange text, and data analysis, reporting, and publication tasks are in purple text. Bars for each task are color coded according to status of the task. Tasks that fall in the 'Needs to be updated' category are completed in some capacity but there is a significant amount of work to be done to make these tasks operational; these tasks are represented by orange bars. 'Outstanding' tasks have not yet been completed and require the most amount of work to become operational. 'Operational' tasks are currently working but may still require updating and general upkeep. These tasks are represented by blue bars. Each task bar is labeled with the person(s) that play a role in completing the task, where FCL = Field Crew Lead, DMA = Data Manager Assistant, PL = Project Lead, DM = Data Manager, GIS = GIS specialist, and IBP = Institute for Bird Populations.

## 3.2 Task descriptions and Roles and Responsibilities

### 3.2.1 Data collection and preparation

#### 3.2.1.1 *Field Crew Hiring – Project Lead and other IBP staff*

Field crew hiring involves updating position descriptions for Field Crew Lead and Field Crew, distributing position description, review of candidates, interview, selection of staff, and onboarding.

#### 3.2.1.2 *Permitting and reporting – Project Lead and assistant data manager, and GIS Specialist*

Permitting involves obtaining species lists and detections at each park, completing Investigator Annual Reports (IAR), and submitting research permit application at 6 parks. GIS Specialist develops maps of site locations and list of site coordinates to accompany permit application.

#### 3.2.1.3 *Field preparation – Project Lead, assistant data manager, and GIS specialist*

After hiring staff, the Project Lead finds crew accommodations in Flagstaff, purchases field gear for the crew, and conducts online review sessions with field crew, finalizes field schedule and logistics, and print field data collection sheets. Project Lead works with GIS specialist to produce paper and digital maps of survey transects and prepare and update GPS units for field crew. Project Lead works with the assistant data manager to prepare field tablets (IT components and uploading maps and photos) and produce navigational report and task list for the field visits. Project Lead prepares Automated Recording Units (ARU; AudioMoth loggers) for field deployment. The GIS Specialist creates a grid of Acoustic Recording Unit (ARU) site locations based on the criteria set by the Project Lead, develops field maps, and manages the upload and download of coordinate data on GPS units.

Although field data are not entered into the Access database until completion of the field season, the Project Lead works with the assistant data manager before the field season to prepare the Access Bird data entry application for use immediately after the field season ends.

#### 3.2.1.4 *Field training – Project Lead and IBP staff*

Field training is a 3-week, in-person training with field crew (1 lead and 3 technicians) and IBP staff that consists of field visits and review sessions to learn bird songs and habitat classifications, review of field protocol and safety procedures, camping/backpacking training, review of GPS/tablet use. The science communicator trains on outreach and communication with the general public. Field crew is tested on identification skills by sight and sound at the end of training and cannot collect field data unless they pass.

#### 3.2.1.5 *Field visits – Project Lead and field crew*

Prior to training, the Project Lead and IBP staff deploy 3-4 ARU grids at 2 parks (GRCA and PEFO). ARUs are deployed for 3 months and retrieved in early July. Point count surveys take place as 6-8 day survey visits between April 28-June 30. Data collected during the point counts includes distance sampling of bird communities, survey specific environment conditions, habitat characterizations at each point count location, GPS coordinates and cardinal direction photos for newly established survey locations, documented breeding behavior on survey transects, and a species list of birds detected at each park. Data are collected on paper data sheets.

After each park visit, the GIS Specialist downloads the GPS coordinate data and the assistant data manager exports all new site photos for storage on the common drive server.

### 3.2.2 Data processing

#### *3.2.2.1 Field data entered into Access database/Additional data preparation – Project Lead, field crew lead, assistant data manager, and GIS specialist.*

After completing all field surveys, the field crew lead works with the assistant data manager to scan and archive paper data sheets on SCNP common drive server. The discrete field data are entered into the Landbirds database at the completion of the field season. This is done by the bird crew lead using the Access Landbirds Application and with help from the assistant data manager. Verification of the digitized data is performed by the field crew lead after entry of each transect's data. A mechanism exists within the Access database front end for the field crew lead to mark that the digitized data have been verified. The GIS Specialist downloads GPS coordinate data, extracts new point count station sites, standardizes the attribute data, performs QA/QC to identify errors, verifies the accuracy of active site survey locations, appends the data to the geospatial database and works with Project Lead to address any changes to active survey locations that occurred over the field season. Assistant data manager works with the GIS Specialist to update coordinates in the Landbird database. This includes adding coordinates for new points, addressing changes to point names or site/point status (moved or retired locations), and fixing any other errors or issues uncovered. Assistant manager works with Project Lead to update project taxon list based on the American Ornithological Society's checklist (Update annually in September). Assistant data manager uploads point establishment photos from tablets onto the SCNP common drive server. IBP currently processes and stores ARU data.

#### *3.2.2.2 QA/QC and validation of field (discrete) data – Project Lead, assistant data manager, and IBP staff.*

QA/QC of the discrete data should be conducted once a year using Access Landbirds application. QA/QC is the primary responsibility of the Project Lead, although the assistant data manager assists with data preparation and troubleshooting.

#### *3.2.2.3 Discrete data are validated/certified – Project Lead, data manager, assistant data manager, and IBP staff.*

After the Project Lead has completed QA/QC of the data, the data manager will certify the data by ensuring all errors have been accounted for and will upload the validated data onto the Landbirds database on Access.

#### *3.2.2.4 Geospatial data management – GIS Specialist*

The GIS Specialist oversees the management of bird monitoring site locations within an ESRI file geodatabase. The GIS Specialist is responsible for updating site location data, performing QA/QC checks and visual inspections to ensure data quality. Additionally, the GIS Specialist maintains location attribute information for each site and develops metadata documenting spatial data development processes, ensuring data integrity and accessibility.

### 3.2.3 Data analysis, reporting, and publication

#### 3.2.3.1 *Data publication – Project Lead/data manager*

Data publication will occur after data has undergone QA/QC and is validated/certified by the data manager. Metadata for this protocol will be updated and formatted into EML. The data manager will work with the Project Lead to ensure the final data package is properly formatted using EML and data files are in machine-readable format (CSV), which will be provided by the Project Lead. The data manager will run congruence checks on the data package using NPS provided tools and ensure the data files have the same data as the Certified Landbirds Access, excluding any formatting or protected data. The data manager will publish the data package on NPS Data Store after review from the Project Lead.

#### 3.2.3.2 *Data analysis/reporting – TBD, and GIS Specialist*

Reporting on survey effort and species detections are provided to each park as part of the Investigator Annual Reports. Trend reporting occurs as funding permits (SCC – Resource Condition Assessments, Inventories, and other external sources and based funding; e.g., Jones et al 2024, 2025, in press). The GIS Specialist conducts spatial analyses to support reporting efforts, and generates maps for annual reports. Additionally, the GIS Specialist reviews summaries and reports to verify the accuracy of the spatial data and sources, and compiles additional spatial information to enhance analyses and reporting.

### 3.3 Future needs/wants

- Figure out best way to deal with photos (actively working on as of September 2025)
- ARU data (all aspects)
- Assistant data manager will list all steps for data entry preparation and validation.
- Taxonomy (we should explore further)

## 4 Upland vegetation

### 4.1 Ideal Data Life Cycle for Upland Vegetation

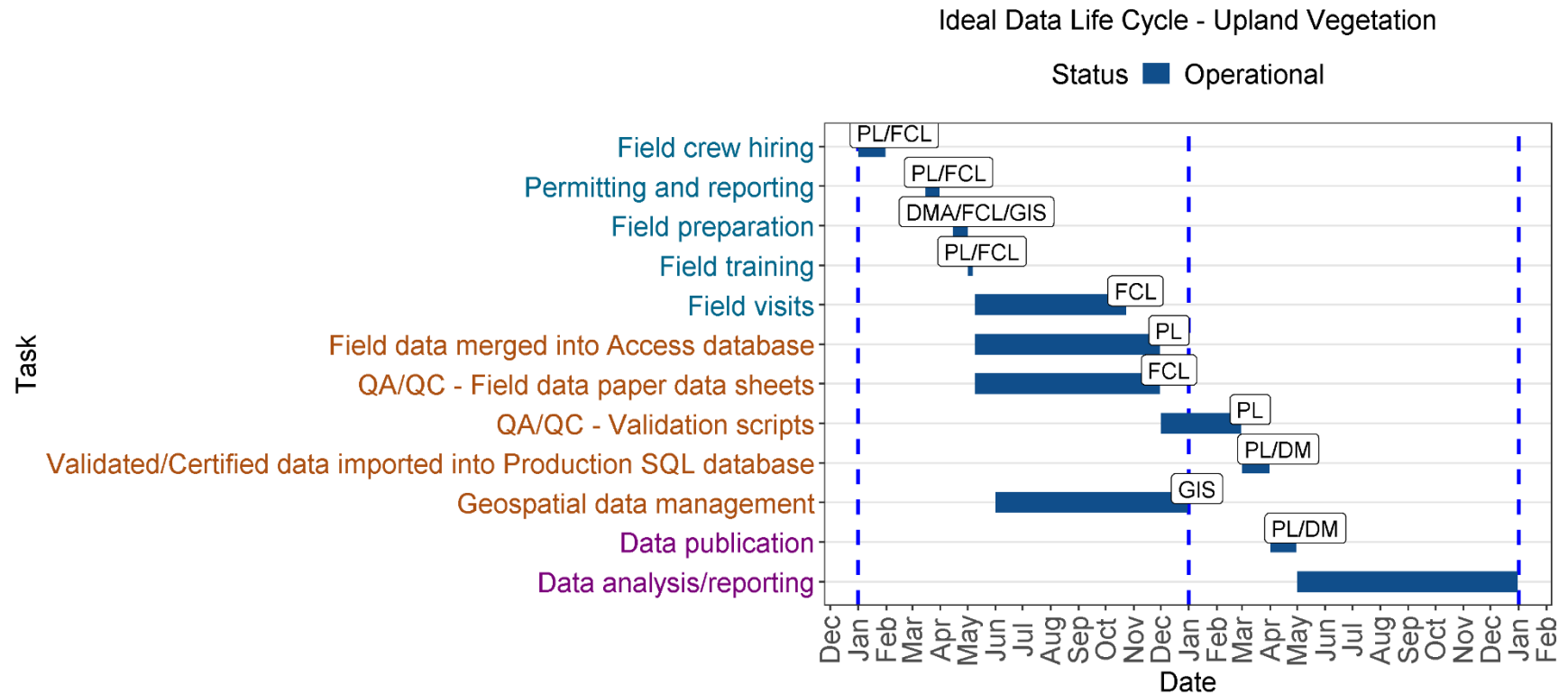


Figure 3.1. This chart is a representation of the workflow for the Upland Vegetation vital sign. Tasks are color coded according to the type of task, where data collection tasks are in blue text, data entry and QA/Qc tasks are in orange text, and data analysis, reporting, and publication tasks are in purple text. Bars for each task are color coded according to status of the task. Tasks that fall in the 'Needs to be updated' category are completed in some capacity but there is a significant amount work to be done to make these tasks operational; these tasks are represented by orange bars. 'Operational' tasks are currently working but may still require updating and general upkeep. These tasks are represented by blue bars. Each task bar is labeled with the person(s) that play a role in completing the task, where FCL = Field Crew Lead, DMA = Data Manager Assistant, PL = Project Lead, and DM = Data Manager.

## 4.2 Task descriptions and Roles and Responsibilities

### 4.2.1 Data collection and preparation

#### 4.2.1.1 *Field Crew Hiring – Project Lead, field crew lead, assistant data manager, and NAU coordinator*

Field crew hiring involves updating position descriptions for field crew, distributing position description, review of candidates, interview, selection of staff, and onboarding. Need to coordinate with NAU on hiring. As soon as hiring selections are made, work with assistant data manager to get new employees carded.

#### 4.2.1.2 *Permitting and reporting – Project Lead and field crew lead*

Permitting involves completing Investigator Annual Reports (IAR) and submitting research permit application at 10 parks.

#### 4.2.1.3 *Field preparation – Project Lead, field crew lead, assistant data manager, and GIS specialist*

The Project Lead purchases field gear for the crew and field supplies, finalizes field schedule and logistics. Project Lead and field crew lead work with GIS specialist to finalize plot list, produce paper and digital maps of plots, prepare and update GPS units for the season, and imports tree maps into ESRI Field Maps on small tablets. Project Lead and field crew lead work with the assistant data manager to prepare field tablets (IT components and uploading maps and photos) and produce revisit reports. Field crew lead prints field revisit plot sheets updates field guides and plant lists. Assistant data manager incorporates any changes, fixes, or updates to the Access field data entry application.

#### 4.2.1.4 *Field training – Project Lead, field crew lead, assistant data manager, and GIS Specialist*

Field training is a 1-week, in-person training with field crew (4 technicians) and NPS staff that consists of field visits and review sessions to learn protocol, safety procedures, plant identification, and GPS/tablet use. The science communicator provides the crew training on outreach and communication with the general public. The GIS specialist trains crew on navigation and use of GPS and ESRI Field Maps application on tablets.

#### 4.2.1.5 *Field visits – Project Lead and field crew*

Data is collected in 8-day sampling hitches between May and October. Data varies by site but includes understory vegetation presence and abundance, soil stability, tree canopy cover/closure, tree basal area density, photopoints, and seeding/sapling abundance.

### 4.2.2 Data processing

#### 4.2.2.1 *Field data merged into Access database – Project Lead*

Access databases from field tablets are merged into single park Access databases after each field visit. This is performed using the Access Upland Vegetation merge tool, which allows the user to merge field tablet Access database. This is primarily done by the Project Lead during the field season once the crew finishes all work at a park.

#### *4.2.2.2 QA/QC - Field data paper data sheets – Field crew lead*

Any paper datasheets generated during the field season are 100% reviewed against the Access database for each park by the field crew lead prior to validation. Plot summary datasheets are also reviewed for changes or errors and updates made to the Access database.

#### *4.2.2.3 QA/QC and validation of field (discrete) data – Project Lead*

QA/QC of the discrete data should be conducted once a year. An R script has been written for this protocol to perform QA/QC of the data. QA/QC is the primarily the responsibility of the Project Lead, although the data manager will assist with any troubleshooting. Once all errors have been accounted for and tables prepared, the validated data ready to be upload into the SCPN\_UplandEvent database on the Production instance of SQL server following the instructions on Teams.

#### *4.2.2.4 Validated/Certified data imported into Production SQL database– Project Lead and data manager*

Following validation, the Project Lead works with the data manager or assistant data manager to import the data into the SCPN\_UplandEvent database. Instructions for this process can be found at S:\SQLServer\zDM\_Processes.

#### *4.2.2.5 QA/QC and Validation of Spatial Data – GIS Specialist*

QA/QC of the spatial data should be conducted at the end of each field season. The GIS Specialist downloads spatial data from GPS units and tablets using the appropriate software. To validate the spatial data and attributes, both automated checks and visual inspections are employed to identify errors and ensure logical consistency. The responsibility for QA/QC primarily falls to the GIS Specialist; however, support may be provided by other team members during the troubleshooting process. Once all errors have been addressed and the tables are properly prepared, the validated data will be appended to the Upland\_AllSites geodatabase.

### *4.2.3 Data analysis, reporting, and publication*

#### *4.2.3.1 Data publication – Project Lead/data manager*

Data publication will occur after data has undergone QA/QC and is validated by the data manager. Metadata for this protocol has been updated and formatted into EML. The data manager will work with the Project Lead to ensure the final data package is properly formatted using EML and data files are in machine-readable format (CSV), which will be provided by the Project Lead. The data manager will certify the data by ensuring that the data package has been verified using NPS provided tools and that data files are the same as data in the Production instance on SQL server, excluding any formatting or protected data. The data manager will publish the data package on NPS Data Store after review from the Project Lead.

#### *4.2.3.2 Data analysis/reporting – Project Lead/data manager, and GIS Specialist*

Annual data summaries are produced through R scripts by March of the year following the field season. They are provided to park staff and uploaded as part of the Research Permit Reporting System. The GIS Specialist performs spatial analyses to support reporting efforts and

generates maps for annual reports. They also review summaries and reports to ensure the accuracy of spatial data and sources, while compiling additional spatial information to enhance analysis and reporting. Need to update this section

#### 4.3 Future needs/wants

- Update validation script
- Annual data summaries are complete but figures are due to be updated.



## 5 Land Surface Phenology

In spring 2025, the project lead for this vital sign retired. The status of the vital sign prior to this retirement is summarized below. At present, the position remains vacant and is not expected to be refilled in the near future.

### 5.1 Ideal Data Life Cycle for Land Surface Phenology

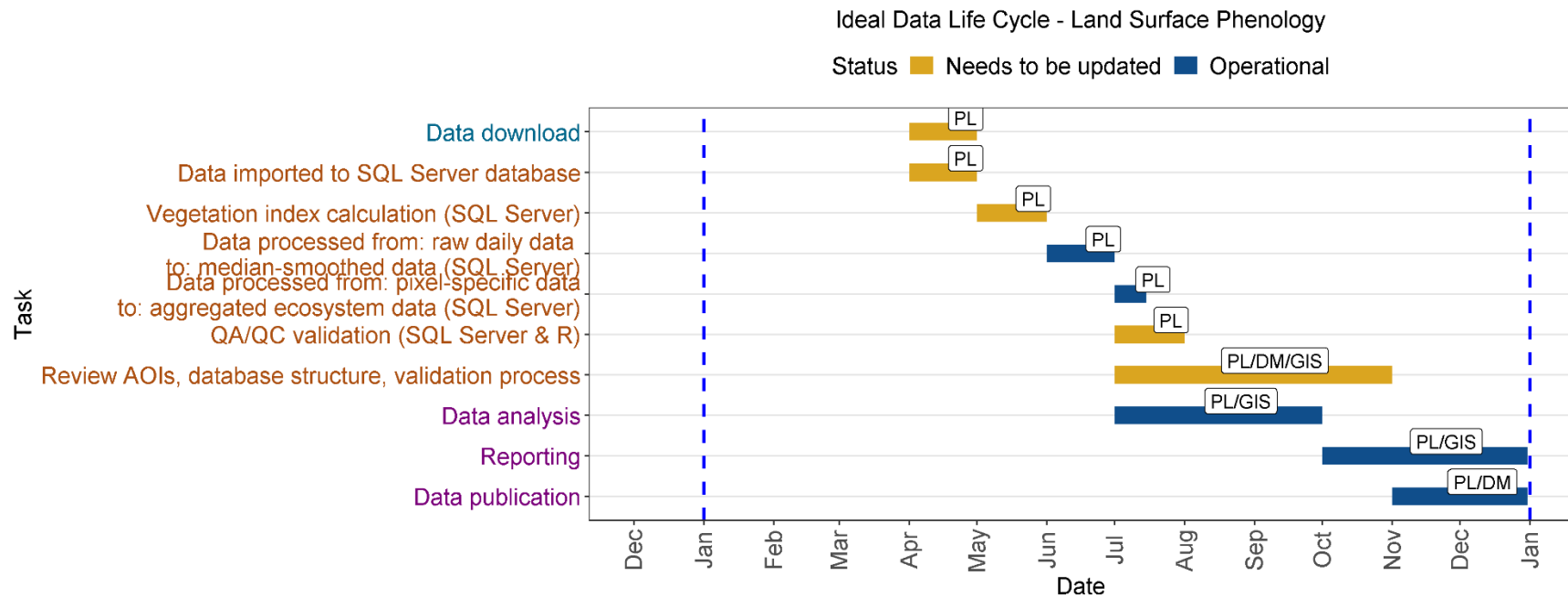


Figure 3.1. This chart is a representation of the workflow for the Land Surface Phenology vital sign. Tasks are color coded according to the type of task, where data download and import tasks are in blue text, data processing and QA/QC tasks are in orange text, and data analysis, reporting, and publication tasks are in purple text. Bars for each task are color coded according to status of the task. Tasks that fall in the 'Needs to be updated' category are represented by orange bars. 'Operational' tasks are currently working but may still require updating and general upkeep. These tasks are represented by blue bars. Each task bar is labeled with the person(s) that play a role in completing the task, where PL = Project Lead, DM = Data Manager, and GIS = GIS Specialist.

## 5.2 Task descriptions and Roles and Responsibilities

The Land Surface Phenology protocol estimates daily vegetation greenness in target ecosystems in SCPN parks at a spatial scale of 250 m pixels. The protocol uses downloads of daily, publicly available, satellite data. The protocol creates pixel-specific daily record sequences of filtered, smoothed data as well as spatially aggregated statistics for target ecosystems. Annual phenological statistics are calculated, and long term patterns are analyzed. Key data products of this protocol are pixel- and ecosystem- specific time series of vegetation indices.

Broad protocol tasks are described here and are accomplished by the protocol manager, data manager, and GIS specialist.

### 5.2.1 Data collection

#### 5.2.1.1 Data download– Project Lead

Data for each area of interest are downloaded using scheduled scripts. The download was previously done through Python scripts to Google Earth Engine. Because of issues with NPS access to Google Earth Engine for production use, scripts are being migrated to download using APPEEARS data provider and either R or a combination of SqlServer TSQL and Powershell. *Modify this line to just reflect the new process once script migration is complete.*

### 5.2.2 Data processing

#### 5.2.2.1 Data imported to SqlServer database – Project Lead

Automated annual data importing to SQLServer is done through TSQL scripts, stored procedures, and job agents. These scripts need to be modified to reflect the shift from EarthEngine / Python downloads to APPEEARS / SqlServer TSQL - Powershell.

#### 5.2.2.2 Vegetation index calculation (SQL Server) – Project Lead

Automated annual vegetation index calculation will be done through existing SQLServer job agents that run TSQL stored procedures. The stored procedures need to be modified to add a step of NDVI calculation and cloud-snow-aerosols-data screening (steps that were previously done using the EarthEngine / Python process but cannot be accomplished in the APPEEARS / Powershell). – Project Lead

#### 5.2.2.3 Data processed *from: raw daily data to: median-smoothed data (SQL Server)*– Project Lead

Automated annual calculation of pixel-specific daily median-smoothed data is done through SQLServer job agents that run TSQL stored procedures.

#### 5.2.2.4 Data processed *from: pixel-specific data to: aggregated ecosystem data (SQL Server)*

Automated annual calculation of aggregated ecosystem daily and annual phenological data is done through SQLServer job agents that run TSQL stored procedures.

#### 5.2.2.5 QA/QC validation (SQL Server & R)

Data are checked to ensure that complete records are present for each pixel without duplicates, that data from both satellites are present for each year, that the median-smoothed record is complete for each pixel, and that ecosystems have ecologically plausible ranges and patterns. Done with R scripts, phenology raster graphs, ecosystem summaries.

#### 5.2.2.6 Review AOIs, database structure, validation process

Review the areas of interest and the data sources and phenological metrics calculated and modify as needed. GIS specialist may assist with creating new areas of interest. Also review the validation process to ensure that it is sufficient.

### 5.2.3 Data analysis, reporting, and publication

#### 5.2.3.1 Data analysis

Data will be analyzed using a combination of trend analysis, analysis of covariates, and spatial analysis and interpreted in the context of existing literature. The GIS Specialist performs spatial analyses to support analysis and reporting efforts

#### 5.2.3.2 Reporting

Reports will be produced using the analysis products and will provide some combination of trend analysis, analysis of covariates, spatial analysis and interpretation in the context of existing literature.

#### 5.2.3.3 Data publication – Project Lead/data manager

Data publication will occur after data has undergone QA/QC and is validated by the data manager. The data manager will work with the Project Lead to ensure the final data package is properly formatted using EML and data files are in machine-readable format (CSV), which will be provided by the Project Lead. The data manager will certify the data by ensuring that the data package has been verified using NPS provided tools and that data files are the same as data on SQL server, excluding any formatting or protected data. The data manager will publish the data package on NPS Data Store after review from the Project Lead.

## 5.3 Future wants/needs

Things that need to happen

- Migration of data source to the VIIRS satellite data
- Migration of data download and VI processing scripts to the APPEARS data repository
- Completion of metadata for SQLServer database
- Migration of data download, scripts, and automated online reporting to regional/national level (this is a side task, not part of vital sign, but most of it will overlap with vital sign tasks).

## 6 Climate data

In spring 2025, the project lead for this vital sign retired. The status of the vital sign prior to this retirement is summarized below. At present, the position remains vacant and is not expected to be refilled in the near future. During 2026, SCPN will continue to provide periodic maintenance and fund data acquisition and processing for three climate stations in three NPS units (NAVA, PEFO, WUPA) while each park decides whether to decommission its climate station or assume maintenance and data processing responsibilities thereafter. SCPN will cease supporting climate stations in 2017.

### 6.1 Ideal Data Life Cycle for Climate Data

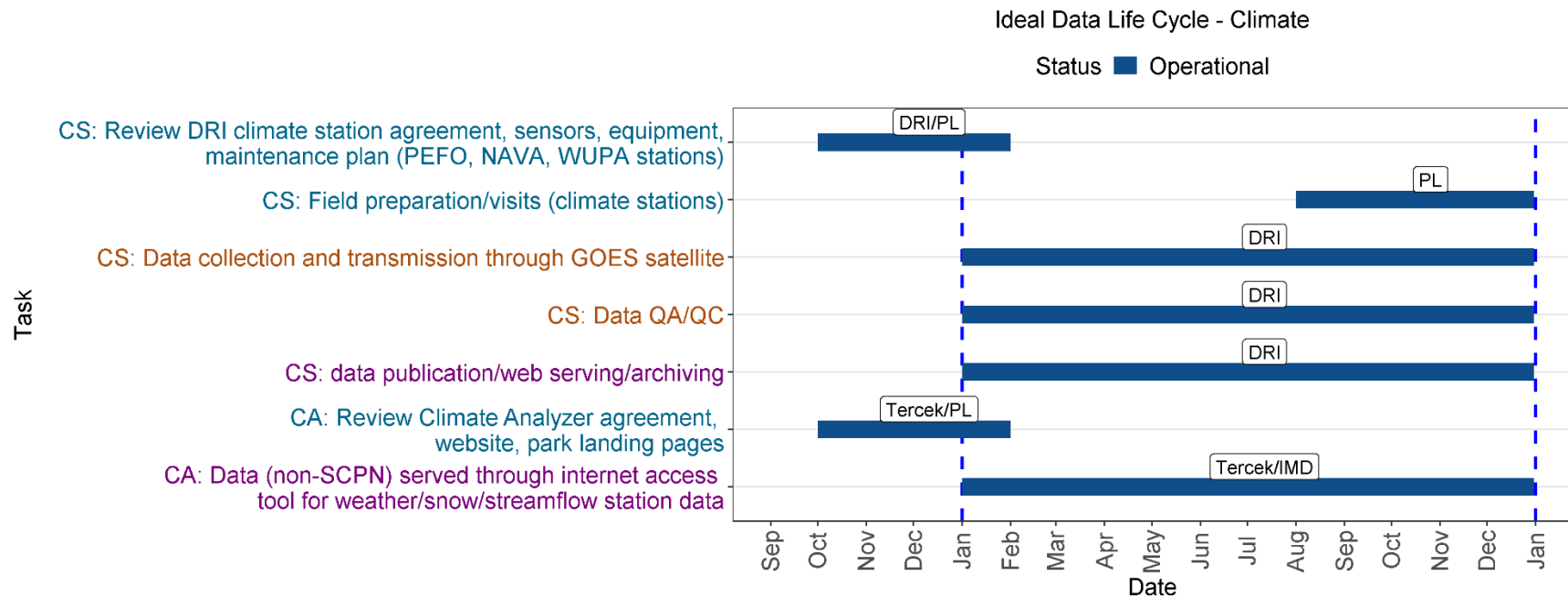


Figure 5.1. This chart is a representation of the workflow for the Climate data vital sign, divided into the climate station related track (task prefix **CS**) and the Climate Analyzer track (task prefix **CA**). Tasks are color coded according to the type of task, where annual preparation and field visits are in blue text, data collection, transmission, processing, and QA/QC tasks are in orange text, and data analysis, reporting, and publication

*tasks are in purple text. Bars for each task are color coded according to status of the task. For the climate vital sign, all tasks are currently 'Operational' and are represented by blue bars. Each task bar is labeled with the person(s) that play a role in completing the task, where PL = Project Lead, DRI = Desert Research Institute, Tercek = Walking Shadow Ecology/Mike Tercek.*

## 6.2 Task descriptions and Roles and Responsibilities

The Climate vital sign consists of two tracks: The Climate Station track and the Climate Analyzer track.

The Climate Station track refers to the collection and publication of weather/climate data at three automated climate monitoring stations that were inherited by the parks (NAVA, PEFO, WUPA) after being installed by NOAA as part of the discontinued Regional Climate Reference Network program. For the CS track, SCPN collaborates with the Desert Research Institute (DRI). In this collaboration, the SCPN completes basic station maintenance (visiting stations to empty precipitation bucket, run calibrations, or troubleshoot under the guidance of DRI staff). The SCPN funds DRI to do station data ingestion, QC, and publication. SCPN has also funded DRI on an as-needed basis to do additional tasks such as installing new equipment (typically purchased by SCPN, but it can also be purchased by the parks). For the data lifecycle, the DRI ingests data transmitted through the GOES satellite network. DRI then has scripted QA/QC scripts and manually reviews discrepancies in the data identified by these scripts. Data are published through the DRI/WRCC website on an hourly basis. These data can then be immediately and automatically used by a variety of other programs that use weather station data across the nation, such as for flood warning, calibration of radar precipitation products, and other uses. SCPN is not responsible for data management or archiving, except indirectly to provide funding by DRI.

The Climate Analyzer track (CT) refers to the creation and maintenance of a web-based climate data access tool called the Climate Analyzer. This tool brings together links, graphs, and tables of information from a variety of climate data sources including multiple weather station networks, streamflow gauges, and snow monitoring sites. Pages have been created that are specific to individual I&M networks and are funded through a multi-network agreement. There are no further data management or archiving tasks in this track because the Climate Analyzer acts only as an aggregator of data published and maintained by other archives and networks of monitoring stations.

### 6.2.1 Annual preparation and field visits (*Climate station track*)

#### 6.2.1.1 *CS: Review DRI climate station agreement, sensors, equipment, maintenance plan (PEFO, NAVA, WUPA stations) – Desert Research Institute and Project Lead*

Each year SCPN pays for the next year of data serving, QC checks and processing. During the period from October to March, SCPN and DRI staff should review any additional maintenance work or equipment or services needed for the coming year.

#### 6.2.1.2 *CS: Field preparation/visits (climate stations) – Project Lead*

Project Lead visits stations 1-2 times each year to empty the precipitation bucket and do any suggested simple maintenance or troubleshooting suggested by DRI staff.

## 6.2.2 Data collection and processing (*Climate station track*)

### 6.2.2.1 CS: Data collection and transmission through GOES satellite – Project Lead and DRI

Data are collected by sensors and transmitted hourly through the GOES satellite network for each station through the HADS system (for example, the Wupatki HADS identifier information page is [https://hads.ncep.noaa.gov/cgi-bin/hads/interactiveDisplays/displayMetaData.pl?table=dc&nesdis\\_id=CD0D4058](https://hads.ncep.noaa.gov/cgi-bin/hads/interactiveDisplays/displayMetaData.pl?table=dc&nesdis_id=CD0D4058) . The DRI downloads the HADS data from NOAA. – Desert Research Institute

### 6.2.2.2 CS: Data QA/QC – DRI

DRI staff are responsible for data QA/QC through an annual payment by SCPN. Data are checked to ensure that complete records are present and that data patterns are consistent with expected local patterns and with multiple sensors (when present). DRI notifies SCPN staff of anomalies and advises on checks that can be made at the station.

## 6.2.3 Data analysis, reporting, and publication (*Climate station track*)

### 6.2.3.1 CS: data publication/web serving/archiving – Desert Research Institute

The Desert Research Institute publishes and archives the data through their website and databases.

## 6.2.4 Annual preparations (*Climate Analyzer track*)

### 6.2.4.1 Review Climate Analyzer agreement, website, park landing pages – Tercek-Walking Shadow Ecology / Project Lead

Review any needs/modifications for the climate analyzer. Most years there are none.

## 6.2.5 Data publication (*Climate Analyzer track*)

### 6.2.5.1 Data (non-SCPN) served through internet access tool for weather/snow/streamflow station data – Tercek-Walking Shadow Ecology

Climate Analyzer products are served based on individual networks through <http://www.climateanalyzer.org/> . Because the data are collected, maintained, and officially archived by other entities, SCPN does not have a data management responsibility for these data. The primary users of the Climate Analyzer products are expected to be park staff and cooperators/researchers working in parks.